Yanis Bérard

Student at the École Normale Supérieure de Rennes – Mathematics

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Education

ENS Rennes *2025 – Present*
Student (*élève normalien*), Pure Mathematics track (*Magistère de Mathématiques*).

Université de Rennes *2025 – Present*
Master's degree in Mathematics, first year (M1, 2026 – Present); Bachelor's degree in Mathematics, third year (Licence 3, 2025 – 2026).

Lycée du Parc, Lyon *2024 – 2025*
CPGE MP*1 (two-year undergraduate preparatory classes for the Grandes Écoles).

Lycée La Martinière Monplaisir, Lyon *2022 – 2024*
CPGE MPSI → MP*.

Lycée Honoré d'Urfé, Saint-Étienne *2022*
Baccalauréat, Mathematics and Physics majors, with highest honours (*mention Très Bien*).

Research Experience

Research internship, Institut Camille Jordan, Lyon *May – June 2026*
Supervised by Jean-Yves Welschinger. Report: *Local Holomorphy and Global Algebraicity: The Road to GAGA* (compact Riemann surfaces, covering theory and the Riemann–Hurwitz formula, sheaf cohomology, Riemann–Roch theorem, Serre duality, projective embedding, Chow's theorem, GAGA for curves).

GATE Laboratory, Saint-Étienne *2023*
Study of voting theory and numerical simulations around Arrow's impossibility theorem.

Academic Projects

- **Expository paper (2026):** *An elementary proof of Bézout's theorem on algebraic curves.* Study of intersections of plane projective curves, local intersection multiplicities, explicit examples and geometric interpretation. Three-hour presentation to fellow students.
- **Expository paper:** *The Kakeya conjecture* over finite and local fields, and over the reals.
- **Group theory notes:** simple groups of order < 60; duality for finite abelian groups.
- **MATh.en.JEANS (2022):** recovering a triangle from its special points.

TIPE (supervised research projects in CPGE)

- **Elections: the rules of the democratic game** (2022–23): Arrow's theorem, computational social choice.
- **Heads or tails: paradoxes and random walks** (2023–24): waiting times, Pólya's theorem.
- **Lattices of finite groups and applications** (2024–25): a generation game on finite groups.

Teaching, Outreach & Other Experience

Private tutor in Mathematics and Physics *2021 – Present*
Regular tutoring of students from middle school to preparatory classes (CPGE), including preparation for the competitive entrance exams of the Grandes Écoles.

Outreach: Organisation of the *Rencontres des Jeunes Mathématiciennes et Informaticiennes* (outreach event for young women in mathematics and computer science), design of a workshop.

Mechanical worker – *NTD France*; **Market gardening** – *Ferme Le Bénitier aux Oiseaux*

Skills

- **Languages:** French (native), English (C1), Portuguese (B2), Spanish (B1), Italian (B1).
- **Computer skills:** LaTeX, Python, OCaml.
- **Interests:** Climbing (Loire junior champion), skiing, running, cycling, chess.